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Link ChatGPT: <https://chatgpt.com/share/692aedd9-38cc-800a-820e-79c22ba12fe6>

Link Github : <https://github.com/dinarahmadita06/kuliah-multimedia-25-26/tree/a15cc670e89831b32fb1aa91f3c59190d4a2bbfe/tugas-rPPG>

Pembeda:

- ROI di area dahi (jidat): strip sempit di bagian atas bounding box wajah ( $\approx 20\%$  tinggi,  $\approx 40\%$  lebar) untuk mengurangi artefak hidung/mata.
- Segmentasi kulit YCrCb: threshold kulit + morfologi untuk menstabilkan rata-rata RGB dari ROI.
- Metode POS untuk ekstraksi pulse
- MediaPipe face detection (dengan fallback OpenCV)
- Hybrid estimasi BPM: kombinasi FFT (60%) + peak detection (40%) dengan interpolasi parabola pada FFT.
- Buffer threshold: mulai estimasi BPM saat buffer  $\geq 60\%$  dari 300 frame agar spektrum lebih stabil.

## 1. Import Libraries

**Penjelasan:** Cell ini mengimport semua library yang diperlukan untuk sistem rPPG:

- **cv2** (OpenCV): Untuk akses webcam, manipulasi gambar, dan deteksi wajah
- **numpy**: Untuk operasi array dan komputasi numerik
- **scipy.signal**: Untuk signal processing (filtering, FFT, deteksi peak)
- **matplotlib**: Untuk visualisasi grafik pulse dan spektrum
- **IPython.display**: Untuk update plot secara real-time di Jupyter Notebook
- **deque**: Untuk buffer sliding window yang efisien
- **warnings**: Untuk menyembunyikan warning yang tidak perlu

```
In [1]: import cv2
import numpy as np
import time
from collections import deque
from scipy import signal
from scipy.fft import fft, fftfreq
import matplotlib.pyplot as plt
from IPython.display import display, clear_output
from ipywidgets import Output
import warnings
```

*# Untuk Jupyter Notebook - inline plot*

```
%matplotlib inline

# Suppress warnings
warnings.filterwarnings('ignore')

print("Libraries imported successfully!")
```

Libraries imported successfully!

## 2. Setup MediaPipe dengan Fallback OpenCV

**Penjelasan:** Cell ini mencoba mengimport MediaPipe untuk deteksi wajah. Jika MediaPipe tidak tersedia (belum diinstall), sistem otomatis beralih ke OpenCV Haar Cascade sebagai fallback.

```
In [2]: # MediaPipe dengan fallback ke OpenCV
try:
    import mediapipe as mp
    MEDIAPIPE_AVAILABLE = True
    print("MediaPipe tersedia - akan digunakan untuk deteksi wajah")
except (ImportError, ModuleNotFoundError):
    import types
    MEDIAPIPE_AVAILABLE = False
    print("MediaPipe tidak tersedia - menggunakan OpenCV Haar Cascade")

class _DummyFaceDetection:
    def __init__(self, min_detection_confidence=0.7, model_selection=0):
        self.detector = cv2.CascadeClassifier(
            cv2.data.harcascades + "haarcascade_frontalface_default.xml"
        )

    def process(self, rgb_frame):
        gray = cv2.cvtColor(rgb_frame, cv2.COLOR_RGB2GRAY)
        faces = self.detector.detectMultiScale(gray, 1.1, 5, minSize=(30, 30))
        detections = []
        h, w = gray.shape
        for (x, y, fw, fh) in faces:
            bbox = types.SimpleNamespace(xmin=x/w, ymin=y/h, width=fw/w, height=fh/h)
            loc = types.SimpleNamespace(relative_bounding_box=bbox)
            detections.append(types.SimpleNamespace(location_data=loc))
        return types.SimpleNamespace(detections=detections)

mp = types.SimpleNamespace(
    solutions=types.SimpleNamespace(
        face_detection=types.SimpleNamespace(FaceDetection=_DummyFaceDetection)
    )
)
```

MediaPipe tersedia - akan digunakan untuk deteksi wajah

## 3. Class rPPGDetector

**Penjelasan:** Class utama yang mengimplementasikan seluruh pipeline deteksi detak jantung dengan metode rPPG (remote PhotoPlethysmoGraphy).

**Fungsi-fungsi utama:**

## Inisialisasi ( `__init__` )

- Setup buffer untuk menyimpan 300 frame (window sliding)
- Konfigurasi filter bandpass (50-150 BPM)
- Inisialisasi detector wajah (MediaPipe/OpenCV)

## Deteksi Wajah ( `detect_face_roi` )

- Deteksi wajah dari frame webcam
- Ekstrak ROI (Region of Interest) di area dahi (10-40% dari tinggi wajah)
- Dahi dipilih karena tipis kulitnya dan banyak pembuluh darah

## Segmentasi Kulit ( `skin_segmentation` )

- Konversi BGR → YCrCb color space
- Filter pixel kulit menggunakan threshold warna
- Morfologi (opening & closing) untuk bersihkan noise

## Ekstraksi Sinyal RGB ( `extract_rgb_signal` )

- Ambil rata-rata nilai R, G, B dari ROI dengan mask kulit
- Return tuple (R\_mean, G\_mean, B\_mean)

## Metode POS ( `pos_method` )

- **Plane-Orthogonal-to-Skin:** ekstraksi sinyal pulse dari RGB
- Menggunakan rumus :  $\text{pulse} = G' - (R' + B')/2$
- Tetap bekerja dengan baik meskipun kondisi cahayanya berubah

## Signal Processing

- `detrend_signal` : Hilangkan trend low-frequency dengan Savitzky-Golay
- `bandpass_filter` : Filter Butterworth orde 4 (0.83-2.5 Hz = 50-150 BPM)

## Estimasi BPM

- `estimate_bpm_fft` : Gunakan FFT untuk cari frekuensi dominan + interpolasi parabola
- `estimate_bpm_peaks` : Hitung jarak antar puncak sinyal → konversi ke BPM

## Validasi ( validate\_bpm )

- Reject BPM di luar range 45-160
- Limit perubahan maksimal  $\pm 15$  BPM
- Exponential smoothing untuk hasil yang halus

## Process Frame ( process\_frame )

- Fungsi utama yang dipanggil setiap frame
- Pipeline lengkap: deteksi wajah → ekstrak RGB → hitung BPM → tampilkan

```
In [3]: class rPPGDetector:
    def __init__(self, window_size=300, fps=30):
        self.window_size = window_size
        self.fps = fps
        self.actual_fps = fps

        # Buffer untuk sinyal
        self.signal_buffer = deque(maxlen=window_size)
        self.bpm_history = deque(maxlen=5)
        self.fps_timestamps = deque(maxlen=30)
        self.last_bpm = None

        # Frame tracking untuk monitoring
        self.frame_count = 0
        self.last_print_time = 0
        self.print_interval = 1.0
        self.print_mode = 'newline'
        self.monitoring_log = "" # Untuk widget output
        self.has_new_log = False # Flag untuk log baru

        # Deteksi wajah
        self.mp_face_detection = mp.solutions.face_detection
        self.face_detection = self.mp_face_detection.FaceDetection(
            min_detection_confidence=0.7, model_selection=0
        )

        # Konfigurasi filter (range 50-150 BPM)
        self.lowcut = 0.83
        self.highcut = 2.5
        self.max_bpm_change = 15

        print(f"rPPG Detector diinisialisasi")
        print(f"Deteksi wajah: {'MediaPipe' if MEDIAPIPE_AVAILABLE else 'OpenCV Haa")
        print(f"Window: {window_size} frames ({window_size/fps:.1f}s @ {fps} FPS)")
        print(f"Range BPM: {self.lowcut*60:.0f}-{self.highcut*60:.0f}")

    def detect_face_roi(self, frame):
        """Deteksi wajah dan ekstrak ROI area dahi"""
        rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        results = self.face_detection.process(rgb_frame)

        if not results.detections:
```

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        return None, None

    detection = results.detections[0]
    bboxC = detection.location_data.relative_bounding_box
    h, w, _ = frame.shape

    x = int(bboxC.xmin * w)
    y = int(bboxC.ymin * h)
    width = int(bboxC.width * w)
    height = int(bboxC.height * h)

    # Ekstrak area dahi - BENAR-BENAR DI JIDAT SEKARANG!
    roi_y = max(0, y) # Langsung dari atas detection box (0%, bukan 5%)
    roi_height = min(h - roi_y, int(height * 0.20)) # Cuma 20% tinggi wajah (L
    roi_x = max(0, x + int(width * 0.30)) # Lebih center lagi
    roi_width = min(w - roi_x, int(width * 0.40)) # Lebih sempit: 40% lebar wa

    roi = frame[roi_y:roi_y+roi_height, roi_x:roi_x+roi_width]
    bbox = (roi_x, roi_y, roi_width, roi_height)

    return roi, bbox

def skin_segmentation(self, roi):
    """Segmentasi kulit menggunakan YCrCb color space"""
    ycrcb = cv2.cvtColor(roi, cv2.COLOR_BGR2YCrCb)
    lower = np.array([0, 133, 77], dtype=np.uint8)
    upper = np.array([255, 173, 127], dtype=np.uint8)
    mask = cv2.inRange(ycrcb, lower, upper)

    kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
    mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
    mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)

    return mask

def extract_rgb_signal(self, roi):
    """Ekstrak rata-rata nilai RGB dari ROI dengan mask kulit"""
    if roi is None or roi.size == 0:
        return None

    skin_mask = self.skin_segmentation(roi)

    if cv2.countNonZero(skin_mask) < 100:
        skin_mask = np.ones(roi.shape[:2], dtype=np.uint8) * 255

    b, g, r = cv2.split(roi)
    r_mean = cv2.mean(r, mask=skin_mask)[0]
    g_mean = cv2.mean(g, mask=skin_mask)[0]
    b_mean = cv2.mean(b, mask=skin_mask)[0]

    return (r_mean, g_mean, b_mean)

def pos_method(self, rgb_signals):
    """Metode Plane-Orthogonal-to-Skin untuk ekstraksi pulse"""
    rgb_signals = np.array(rgb_signals)
    mean_rgb = np.mean(rgb_signals, axis=0)

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normalized = rgb_signals / mean_rgb
diff = np.diff(normalized, axis=0)
pulse = diff[:, 1] - (diff[:, 0] + diff[:, 2]) / 2
return pulse

def bandpass_filter(self, data):
    """Bandpass filter untuk isolasi sinyal detak jantung"""
    if len(data) < 64:
        return data

    nyquist = self.fps / 2
    low = max(0.01, min(self.lowcut / nyquist, 0.99))
    high = max(low + 0.01, min(self.hicut / nyquist, 0.99))

    b, a = signal.butter(4, [low, high], btype='band')
    return signal.filtfilt(b, a, data)

def detrend_signal(self, data):
    """Hilangkan trend low-frequency dengan Savitzky-Golay filter"""
    window = min(len(data) // 4, 30)
    if window < 3:
        return data
    window = window if window % 2 == 1 else window + 1
    trend = signal.savgol_filter(data, window, 2)
    return data - trend

def estimate_bpm_fft(self, pulse_signal):
    """Estimasi BPM menggunakan FFT"""
    n = len(pulse_signal)

    # Aplikasikan windowing untuk mengurangi spectral leakage
    window = np.hamming(n)
    windowed_signal = pulse_signal * window

    yf = fft(windowed_signal)
    xf = fftfreq(n, 1/self.actual_fps)

    positive_freqs = xf[:n//2]
    power = np.abs(yf[:n//2]) ** 2

    valid_idx = (positive_freqs >= self.lowcut) & (positive_freqs <= self.hicut)
    valid_freqs = positive_freqs[valid_idx]
    valid_power = power[valid_idx]

    if len(valid_power) == 0:
        return 0, positive_freqs, power

    max_idx = np.argmax(valid_power)

    # Interpolasi parabola untuk akurasi sub-bin
    if 0 < max_idx < len(valid_power) - 1:
        alpha = valid_power[max_idx - 1]
        beta = valid_power[max_idx]
        gamma = valid_power[max_idx + 1]
        p = 0.5 * (alpha - gamma) / (alpha - 2*beta + gamma)
        dominant_freq = valid_freqs[max_idx] + p * (valid_freqs[1] - valid_freqs[0])

```

```

        else:
            dominant_freq = valid_freqs[max_idx]

        return dominant_freq * 60, positive_freqs, power

def estimate_bpm_peaks(self, pulse_signal):
    """Estimasi BPM dengan deteksi puncak sinyal"""
    min_distance = int(self.actual_fps * 0.5)
    peaks, _ = signal.find_peaks(
        pulse_signal,
        distance=min_distance,
        prominence=np.std(pulse_signal) * 0.3
    )

    if len(peaks) < 2:
        return 0

    peak_intervals = np.diff(peaks)
    median_interval = np.median(peak_intervals)
    return (self.actual_fps / median_interval) * 60

def update_fps(self, timestamp):
    """Update FPS aktual berdasarkan timestamp"""
    self.fps_timestamps.append(timestamp)
    if len(self.fps_timestamps) >= 2:
        time_diff = self.fps_timestamps[-1] - self.fps_timestamps[0]
        if time_diff > 0:
            self.actual_fps = (len(self.fps_timestamps) - 1) / time_diff
            self.fps = self.actual_fps

def validate_bpm(self, new_bpm):
    """Validasi dan smoothing BPM"""
    if new_bpm < 45 or new_bpm > 160:
        return 0

    if self.last_bpm is None:
        self.last_bpm = new_bpm
        return new_bpm

    bpm_change = abs(new_bpm - self.last_bpm)

    if bpm_change > self.max_bpm_change:
        validated = self.last_bpm * 0.7 + new_bpm * 0.3
    else:
        validated = self.last_bpm * 0.4 + new_bpm * 0.6

    self.last_bpm = validated
    return validated

def process_frame(self, frame, timestamp):
    """Proses satu frame untuk deteksi BPM"""
    self.update_fps(timestamp)
    self.frame_count += 1

    roi, bbox = self.detect_face_roi(frame)

```

```

if roi is None:
    cv2.putText(frame, "Wajah tidak terdeteksi", (10, 30),
                 cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
    return frame, 0

rgb_values = self.extract_rgb_signal(roi)
if rgb_values is None:
    return frame, 0

self.signal_buffer.append(rgb_values)

# Gambar ROI
x, y, w, h = bbox
cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
cv2.putText(frame, "ROI", (x, y-10),
             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 1)

bpm = 0
bpm_fft = 0
bpm_peaks = 0

if len(self.signal_buffer) >= int(self.window_size * 0.6):
    rgb_array = np.array(list(self.signal_buffer))
    pulse_signal = self.pos_method(rgb_array)
    pulse_signal = self.detrend_signal(pulse_signal)
    pulse_signal = self.bandpass_filter(pulse_signal)

    bpm_fft, _, _ = self.estimate_bpm_fft(pulse_signal)
    bpm_peaks = self.estimate_bpm_peaks(pulse_signal)

    if 50 < bpm_fft < 150 and 50 < bpm_peaks < 150:
        bpm = bpm_fft * 0.6 + bpm_peaks * 0.4
    elif 50 < bpm_fft < 150:
        bpm = bpm_fft
    elif 50 < bpm_peaks < 150:
        bpm = bpm_peaks

    if bpm > 0:
        bpm = self.validate_bpm(bpm)
        self.bpm_history.append(bpm)
        bpm = np.mean(list(self.bpm_history))

# Tampilan
buffer_pct = (len(self.signal_buffer) / self.window_size) * 100
status_color = (0, 255, 0) if buffer_pct >= 60 else (0, 165, 255)

cv2.rectangle(frame, (10, 50), (10 + int(buffer_pct * 3), 70), status_color)
cv2.rectangle(frame, (10, 50), (310, 70), (255, 255, 255), 2)
cv2.putText(frame, f"Buffer: {buffer_pct:.0f}%", (10, 45),
             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)

# Info debug
cv2.putText(frame, f"FPS: {self.actual_fps:.1f}", (10, 150),
             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)
cv2.putText(frame, f"Frames: {len(self.signal_buffer)}/{self.window_size}",
             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)

```

```

if bpm_fft > 0 or bpm_peaks > 0:
    cv2.putText(frame, f"FFT: {bpm_fft:.1f} | Peaks: {bpm_peaks:.1f}", (10,
        cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)

if bpm > 0:
    color = (0, 255, 0) if 50 < bpm < 100 else (0, 165, 255)
    cv2.putText(frame, f"Detak Jantung: {bpm:.1f} BPM", (10, 100),
        cv2.FONT_HERSHEY_SIMPLEX, 1, color, 2)
else:
    cv2.putText(frame, "Menghitung...", (10, 100),
        cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 255), 2)

# Console monitoring - UPDATE FLAG
if timestamp - self.last_print_time >= self.print_interval:
    self.monitoring_log = self.print_monitoring_info(bpm)
    self.has_new_log = True # Flag ada log baru
    self.last_print_time = timestamp
else:
    self.has_new_log = False

return frame, bpm

def print_monitoring_info(self, bpm):
    """Print monitoring info ke console"""
    buffer_size = len(self.signal_buffer)

    if buffer_size >= self.window_size:
        frame_start = self.frame_count - self.window_size + 1
        frame_end = self.frame_count
    else:
        frame_start = 1
        frame_end = self.frame_count

    window_duration = buffer_size / self.actual_fps if self.actual_fps > 0 else
    bpm_str = f"{bpm:.1f}" if bpm > 0 else "---"

    # Return string instead of print - untuk widget output
    return (f"[Frame {frame_start:04d}-{frame_end:04d}] "
        f"Buffer: {buffer_size:3d}/{self.window_size} | "
        f"Window: {window_duration:4.1f}s | "
        f"FPS: {self.actual_fps:5.1f} | "
        f"BPM: {bpm_str:>6s}")

def get_current_signal(self):
    """Ambil sinyal terkini untuk plotting"""
    if len(self.signal_buffer) < self.window_size // 2:
        return None, None, None, None

    rgb_array = np.array(list(self.signal_buffer))
    pulse_signal = self.pos_method(rgb_array)
    pulse_signal = self.detrend_signal(pulse_signal)
    pulse_signal = self.bandpass_filter(pulse_signal)

    _, freqs, power = self.estimate_bpm_fft(pulse_signal)
    time_array = np.arange(len(pulse_signal)) / self.fps

```

```

        return time_array, pulse_signal, freqs, power

print("Class rPPGDetector berhasil didefinisikan!")

```

Class rPPGDetector berhasil didefinisikan!

## 5. Jalankan Program

### Bagian yang Real-time (Berubah-ubah):

1. Window webcam OpenCV - update setiap frame
2. Plot di cell output - **BERUBAH-UBAH** setiap 10 frame (pakai `clear_output()`)
3. Rolling window buffer - 300 frame yang terus bergeser
4. BPM value - update setiap detik

#### Cara Kerja Plot Real-time:

- Setiap 10 frame → `clear_output()` hapus plot lama → tampilkan plot baru
- Jadi plot tidak statis, tapi berubah-ubah sesuai data terbaru!

```

In [4]: def run_rppg_realtime(show_plot=True, update_interval=10):

    detector = rPPGDetector(window_size=300, fps=30)

    cap = cv2.VideoCapture(0)
    cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
    cap.set(cv2.CAP_PROP_FPS, 30)

    if not cap.isOpened():
        print("Error: Tidak dapat membuka webcam")
        return []

    print("SISTEM rPPG REAL-TIME - ROLLING WINDOW MONITORING")
    print("Tekan 'q' untuk keluar, 's' untuk screenshot")
    if show_plot:
        print(f"Plot akan UPDATE/BERUBAH setiap {update_interval} frame")
        print("="*80 + "\n")

    # BUAT 2 OUTPUT AREA TERPISAH - INI KUNCINYA!
    log_output = Output()
    plot_output = Output()

    # Tampilkan kedua area
    display(log_output)
    display(plot_output)

    frame_count = 0
    fps_time = time.time()
    fps_value = 0

    # KUMPULKAN SEMUA LOG AGAR MASUK SAAT EXPORT HTML

```

```

all_logs = []

try:
    while True:
        ret, frame = cap.read()
        if not ret:
            break

        current_timestamp = time.time()
        processed_frame, bpm = detector.process_frame(frame, current_timestamp)

        # PRINT LOG KE LOG_OUTPUT AREA - HANYA KALAU ADA LOG BARU!
        if hasattr(detector, 'has_new_log') and detector.has_new_log:
            with log_output:
                print(detector.monitoring_log)
                # Simpan juga ke list agar bisa di-print ulang di akhir
                all_logs.append(detector.monitoring_log)

        frame_count += 1
        if frame_count % 30 == 0:
            fps_value = 30 / (time.time() - fps_time)
            fps_time = time.time()

        cv2.putText(processed_frame, f"Display FPS: {fps_value:.1f}", (10, 130),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)

        cv2.imshow('Deteksi Detak Jantung rPPG', processed_frame)

        # UPDATE PLOT SECARA REAL-TIME DI JUPYTER
        if show_plot and frame_count % update_interval == 0:
            time_arr, pulse, freqs, power = detector.get_current_signal()

            if pulse is not None:
                # CLEAR HANYA PLOT OUTPUT - BUKAN LOG!
                with plot_output:
                    clear_output(wait=True)

                # Buat plot BARU
                fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 8))
                fig.suptitle(f'Analisis Sinyal rPPG - Frame {frame_count}',
                            fontsize=14, fontweight='bold')

                # Plot 1: Sinyal Pulse
                ax1.plot(time_arr, pulse, 'b-', linewidth=1.5, alpha=0.8)
                ax1.set_xlabel('Waktu (detik)', fontsize=10)
                ax1.set_ylabel('Amplitudo', fontsize=10)
                ax1.set_title('Sinyal Pulse (Metode POS)', fontsize=11)
                ax1.grid(True, alpha=0.3, linestyle='--')
                ax1.axhline(y=0, color='k', linestyle='--', linewidth=0.5, alpha=0.5)

                # Plot 2: Spektrum Frekuensi
                freq_bpm = freqs * 60
                valid_idx = (freq_bpm >= 40) & (freq_bpm <= 200)
                ax2.plot(freq_bpm[valid_idx], power[valid_idx], 'r-', linewidth=1.5, alpha=0.8)
                ax2.fill_between(freq_bpm[valid_idx], power[valid_idx], alpha=0.3)
                ax2.set_xlabel('Detak Jantung (BPM)', fontsize=10)

```

```

ax2.set_ylabel('Power', fontsize=10)
ax2.set_title('Spektrum Frekuensi', fontsize=11)
ax2.grid(True, alpha=0.3, linestyle='--')

if bpm > 0:
    ax2.axvline(bpm, color='g', linestyle='--', linewidth=2
                label=f' Terdeteksi: {bpm:.1f} BPM')
    ax2.legend(loc='upper right', fontsize=10)

    # Info di bawah plot
    info_text = f"Frame: {frame_count} | FPS: {detector.act
fig.text(0.5, 0.02, info_text, ha='center', fontsize=10
        bbox=dict(boxstyle='round', facecolor='wheat',

plt.tight_layout(rect=[0, 0.03, 1, 0.97])

# Simpan figure agar muncul di export HTML
try:
    fig.savefig('rppg_plot_latest.png', dpi=150)
except Exception:
    pass

# Display plot di plot_output
display(fig)
plt.close(fig) # Tutup untuk hemat memory

key = cv2.waitKey(1) & 0xFF
if key == ord('q'):
    print("Keluar dari program...")
    break
elif key == ord('s'):
    filename = f'screenshot_{int(time.time())}.png'
    cv2.imwrite(filename, processed_frame)
    print(f"Screenshot tersimpan: {filename}")

except KeyboardInterrupt:
    print("Dihentikan oleh user")

finally:
    cap.release()
    cv2.destroyAllWindows()
    plt.close('all')
    print("Program selesai")

# Return semua log agar bisa diprint di cell berikutnya (masuk saat export)
return all_logs

print("Fungsi run_rppg_realtime berhasil didefinisikan!")
print("Plot akan BERUBAH-UBAH secara real-time menggunakan clear_output()!")

```

Fungsi run\_rppg\_realtime berhasil didefinisikan!  
 Plot akan BERUBAH-UBAH secara real-time menggunakan clear\_output()!

---

## Hasil plot

Saat program berjalan, akan muncul 2 grafik yang diperbarui berkala sesuai data terbaru:

## Grafik 1: Sinyal Pulse (Atas)

**Sumbu X** = Waktu (detik) | **Sumbu Y** = Amplitudo (kecil, biasanya  $\pm 0.002$ )

Contoh (Frame  $\sim 950$ , BPM  $\approx 71.3$ ):

- Awal (detik 0–3) bisa kurang stabil → adaptasi pencahayaan.
- Setelah stabil, gelombang berulang dengan interval  $\sim 0.8$  detik →  $60 / 0.8 \approx 75$  BPM.
- Amplitudo konsisten kecil dan tidak drift jauh dari garis nol.

## Grafik 2: Spektrum Frekuensi (Bawah)

**Sumbu X** = BPM (40–200) | **Sumbu Y** = Power

Interpretasi contoh:

- Satu puncak dominan dan tajam di kisaran 70–80 BPM → detak utama.
- Puncak lain lebih kecil / melebar → harmonik atau noise.
- Garis hijau putus-putus menandai BPM final ( $\approx 71.3$  BPM).

## Info Box (Contoh)

Frame: 950 | FPS: 20.5 | BPM: 71.3

- **Frame:** Total frame terproses.
- **FPS:** Kecepatan aktual.
- **BPM:** Detak jantung akhir yang sudah difilter & dismoothing.

```
In [8]: # Jalankan sistem rPPG dengan plot yang BERUBAH-UBAH!
# Plot akan di-UPDATE setiap 10 frame di output cell ini

logs = run_rppg_realtime(show_plot=True, update_interval=10)

# Cetak ulang semua log agar ikut tersimpan di export HTML
for line in logs:
    print(line)
```

rPPG Detector diinisialisasi  
Deteksi wajah: MediaPipe  
Window: 300 frames (10.0s @ 30 FPS)  
Range BPM: 50-150

SISTEM rPPG REAL-TIME - ROLLING WINDOW MONITORING

Tekan 'q' untuk keluar, 's' untuk screenshot

Plot akan UPDATE/BERUBAH setiap 10 frame

=====

SISTEM rPPG REAL-TIME - ROLLING WINDOW MONITORING

Tekan 'q' untuk keluar, 's' untuk screenshot

Plot akan UPDATE/BERUBAH setiap 10 frame

=====

Output()

Output()

Screenshot tersimpan: screenshot\_1764715781.png

Screenshot tersimpan: screenshot\_1764715782.png

Keluar dari program...

Program selesai

|                   |                 |               |           |            |
|-------------------|-----------------|---------------|-----------|------------|
| [Frame 0001-0001] | Buffer: 1/300   | Window: 0.0s  | FPS: 30.0 | BPM: ---   |
| [Frame 0001-0032] | Buffer: 32/300  | Window: 1.0s  | FPS: 30.6 | BPM: ---   |
| [Frame 0001-0063] | Buffer: 63/300  | Window: 2.1s  | FPS: 30.1 | BPM: ---   |
| [Frame 0001-0093] | Buffer: 93/300  | Window: 3.1s  | FPS: 29.7 | BPM: ---   |
| [Frame 0001-0124] | Buffer: 124/300 | Window: 4.1s  | FPS: 30.1 | BPM: ---   |
| [Frame 0001-0151] | Buffer: 151/300 | Window: 7.2s  | FPS: 20.9 | BPM: ---   |
| [Frame 0001-0170] | Buffer: 170/300 | Window: 10.3s | FPS: 16.5 | BPM: ---   |
| [Frame 0001-0181] | Buffer: 181/300 | Window: 13.6s | FPS: 13.3 | BPM: 102.6 |
| [Frame 0001-0200] | Buffer: 200/300 | Window: 11.6s | FPS: 17.2 | BPM: 98.3  |
| [Frame 0001-0211] | Buffer: 211/300 | Window: 15.5s | FPS: 13.6 | BPM: 98.1  |
| [Frame 0001-0231] | Buffer: 231/300 | Window: 14.8s | FPS: 15.6 | BPM: 101.6 |
| [Frame 0001-0251] | Buffer: 251/300 | Window: 16.4s | FPS: 15.3 | BPM: 106.7 |
| [Frame 0001-0271] | Buffer: 271/300 | Window: 19.9s | FPS: 13.6 | BPM: 108.1 |
| [Frame 0001-0291] | Buffer: 291/300 | Window: 21.5s | FPS: 13.6 | BPM: 102.7 |
| [Frame 0012-0311] | Buffer: 300/300 | Window: 19.8s | FPS: 15.2 | BPM: 107.7 |
| [Frame 0032-0331] | Buffer: 300/300 | Window: 19.5s | FPS: 15.4 | BPM: 95.0  |
| [Frame 0052-0351] | Buffer: 300/300 | Window: 19.6s | FPS: 15.3 | BPM: 79.5  |
| [Frame 0072-0371] | Buffer: 300/300 | Window: 19.2s | FPS: 15.7 | BPM: 89.2  |
| [Frame 0092-0391] | Buffer: 300/300 | Window: 19.8s | FPS: 15.2 | BPM: 74.0  |
| [Frame 0112-0411] | Buffer: 300/300 | Window: 19.4s | FPS: 15.4 | BPM: 72.9  |
| [Frame 0132-0431] | Buffer: 300/300 | Window: 19.4s | FPS: 15.5 | BPM: 70.6  |

Program selesai

|                   |                 |               |           |            |
|-------------------|-----------------|---------------|-----------|------------|
| [Frame 0001-0001] | Buffer: 1/300   | Window: 0.0s  | FPS: 30.0 | BPM: ---   |
| [Frame 0001-0032] | Buffer: 32/300  | Window: 1.0s  | FPS: 30.6 | BPM: ---   |
| [Frame 0001-0063] | Buffer: 63/300  | Window: 2.1s  | FPS: 30.1 | BPM: ---   |
| [Frame 0001-0093] | Buffer: 93/300  | Window: 3.1s  | FPS: 29.7 | BPM: ---   |
| [Frame 0001-0124] | Buffer: 124/300 | Window: 4.1s  | FPS: 30.1 | BPM: ---   |
| [Frame 0001-0151] | Buffer: 151/300 | Window: 7.2s  | FPS: 20.9 | BPM: ---   |
| [Frame 0001-0170] | Buffer: 170/300 | Window: 10.3s | FPS: 16.5 | BPM: ---   |
| [Frame 0001-0181] | Buffer: 181/300 | Window: 13.6s | FPS: 13.3 | BPM: 102.6 |
| [Frame 0001-0200] | Buffer: 200/300 | Window: 11.6s | FPS: 17.2 | BPM: 98.3  |
| [Frame 0001-0211] | Buffer: 211/300 | Window: 15.5s | FPS: 13.6 | BPM: 98.1  |
| [Frame 0001-0231] | Buffer: 231/300 | Window: 14.8s | FPS: 15.6 | BPM: 101.6 |
| [Frame 0001-0251] | Buffer: 251/300 | Window: 16.4s | FPS: 15.3 | BPM: 106.7 |
| [Frame 0001-0271] | Buffer: 271/300 | Window: 19.9s | FPS: 13.6 | BPM: 108.1 |
| [Frame 0001-0291] | Buffer: 291/300 | Window: 21.5s | FPS: 13.6 | BPM: 102.7 |
| [Frame 0012-0311] | Buffer: 300/300 | Window: 19.8s | FPS: 15.2 | BPM: 107.7 |
| [Frame 0032-0331] | Buffer: 300/300 | Window: 19.5s | FPS: 15.4 | BPM: 95.0  |
| [Frame 0052-0351] | Buffer: 300/300 | Window: 19.6s | FPS: 15.3 | BPM: 79.5  |
| [Frame 0072-0371] | Buffer: 300/300 | Window: 19.2s | FPS: 15.7 | BPM: 89.2  |
| [Frame 0092-0391] | Buffer: 300/300 | Window: 19.8s | FPS: 15.2 | BPM: 74.0  |
| [Frame 0112-0411] | Buffer: 300/300 | Window: 19.4s | FPS: 15.4 | BPM: 72.9  |
| [Frame 0132-0431] | Buffer: 300/300 | Window: 19.4s | FPS: 15.5 | BPM: 70.6  |

```
In [9]: # Tampilkan gambar plot
try:
    from IPython.display import Image, display
    display(Image('rppg_plot_latest.png'))
except Exception as e:
    print('Plot terbaru belum tersedia atau gagal ditampilkan:', e)

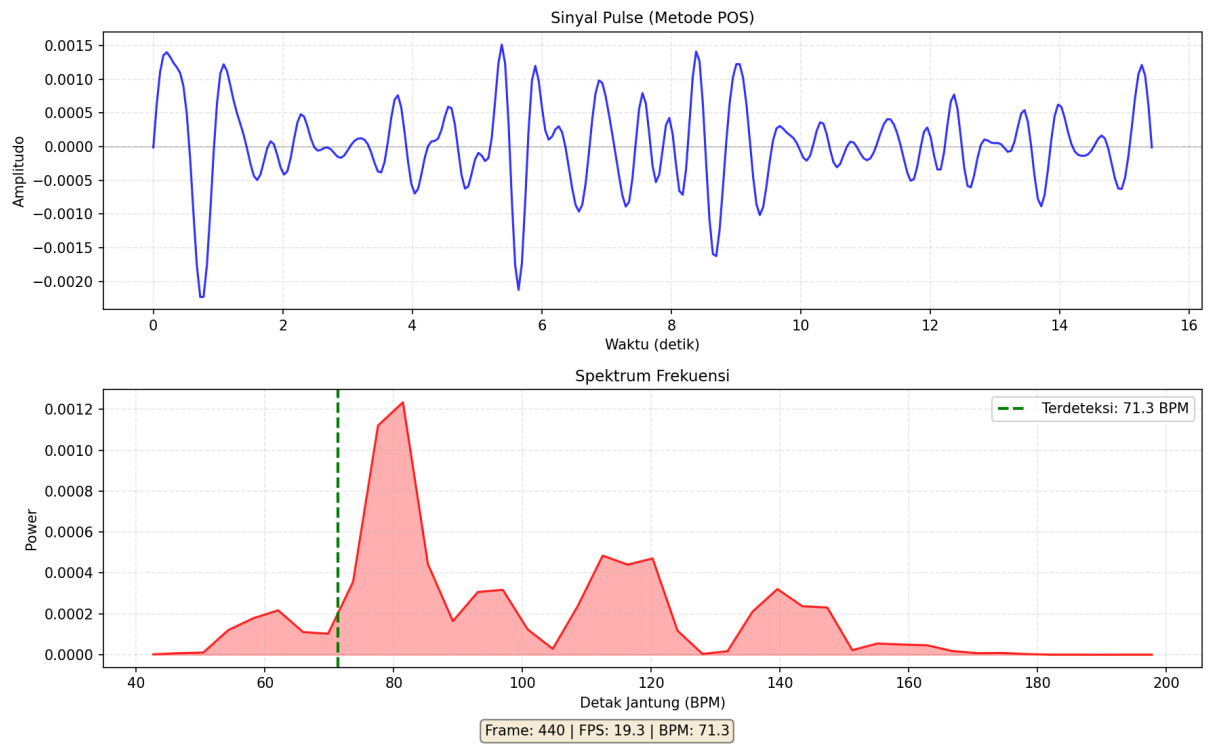
# Opsional: tampilkan screenshot terbaru jika ada
import glob, os
screenshots = sorted(glob.glob('screenshot_*.png'))
```

```

if screenshots:
    latest_ss = screenshots[-1]
    print('Menampilkan screenshot terbaru:', latest_ss)
    try:
        display(Image(latest_ss))
    except Exception as e:
        print('Gagal menampilkan screenshot:', e)
else:
    print('Belum ada file screenshot_*.png')

```

### Analisis Sinyal rPPG - Frame 440



Menampilkan screenshot terbaru: screenshot\_1764715782.png



## 7. Penjelasan Metode - Teori di Balik rPPG

---

### 7.1 Metode POS (Plane-Orthogonal-to-Skin)

**Konsep:** Metode untuk ekstraksi sinyal pulse dari nilai RGB yang lebih robust terhadap perubahan pencahayaan.

#### Alasan penggunaan

- Lebih tahan terhadap perubahan pencahayaan ambient
- Mengurangi pengaruh gerakan kepala
- Channel hijau (G) paling sensitif terhadap perubahan darah

### 7.2 Bandpass Filter

**Tujuan:** Filter frekuensi untuk **hanya loloskan** sinyal detak jantung (50-150 BPM) dan **buang** noise di luar range.

#### Spesifikasi:

- **Type:** Butterworth bandpass filter

- **Order:** 4 (trade-off antara kecuraman & stabilitas)
- **Low cutoff:** 0.83 Hz = 50 BPM (detak paling lambat)
- **High cutoff:** 2.5 Hz = 150 BPM (detak paling cepat)

#### Kenapa range ini?

- Detak jantung normal manusia: 60-100 BPM (istirahat)
  - Detak saat aktivitas ringan: 100-150 BPM
  - Di bawah 50 BPM = kemungkinan bukan detak jantung
  - Di atas 150 BPM = kemungkinan noise
- 

## 7.3 Estimasi BPM dengan FFT

**FFT (Fast Fourier Transform):** Mengubah sinyal dari domain waktu → domain frekuensi untuk menemukan frekuensi dominan.

#### Langkah-langkah:

1. **Windowing:** Hamming window untuk kurangi spectral leakage
2. **FFT:** Transform ke domain frekuensi
3. **Cari Puncak:** Frekuensi dengan power tertinggi
4. **Interpolasi Parabola:** Untuk akurasi sub-bin
  - Pakai 3 titik: sebelum, puncak, sesudah
  - Fit parabola → cari puncak sebenarnya
  - Meningkatkan akurasi dari  $\pm 1$  BPM menjadi  $\pm 0.1$  BPM

**Output:** Frekuensi dalam Hz → dikali 60 → BPM

---

## 7.4 Estimasi BPM dengan Peak Detection

**Konsep:** Deteksi puncak sinyal pulse di domain waktu, lalu hitung interval antar puncak.

#### Langkah-langkah:

1. **Find Peaks:** Deteksi puncak dengan kriteria:
    - **distance** : Minimal 0.5 detik antar puncak
    - **prominence** : Minimal 30% std deviasi
  2. **Hitung Interval**
  3. **Konversi ke BPM**
- 

## 7.5 Kombinasi FFT + Peak Detection

**Alasan penggunaan**

- **FFT:** Akurat untuk sinyal bersih, tapi sensitif terhadap noise
- **Peak Detection:** Robust terhadap noise, tapi kurang akurat